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(19) **United States**(12) **Patent Application Publication****Agerup et al.**(10) **Pub. No.: US 2025/0282450 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **HANG-OFF SYSTEM AND METHOD FOR
USE OF HANG-OFF SYSTEM**(52) **U.S. Cl.**CPC **B63B 27/24** (2013.01); **B63B 27/34**
(2013.01)(71) Applicant: **Econnect Energy AS**, Oslo (NO)(72) Inventors: **Jørgen Fredrik Agerup**, Oslo (NO);
David Mikal Knutsen, Vetre (NO)(73) Assignee: **Econnect Energy AS**, Oslo (NO)(21) Appl. No.: **18/861,434**(22) PCT Filed: **May 5, 2023**(86) PCT No.: **PCT/EP2023/062053**

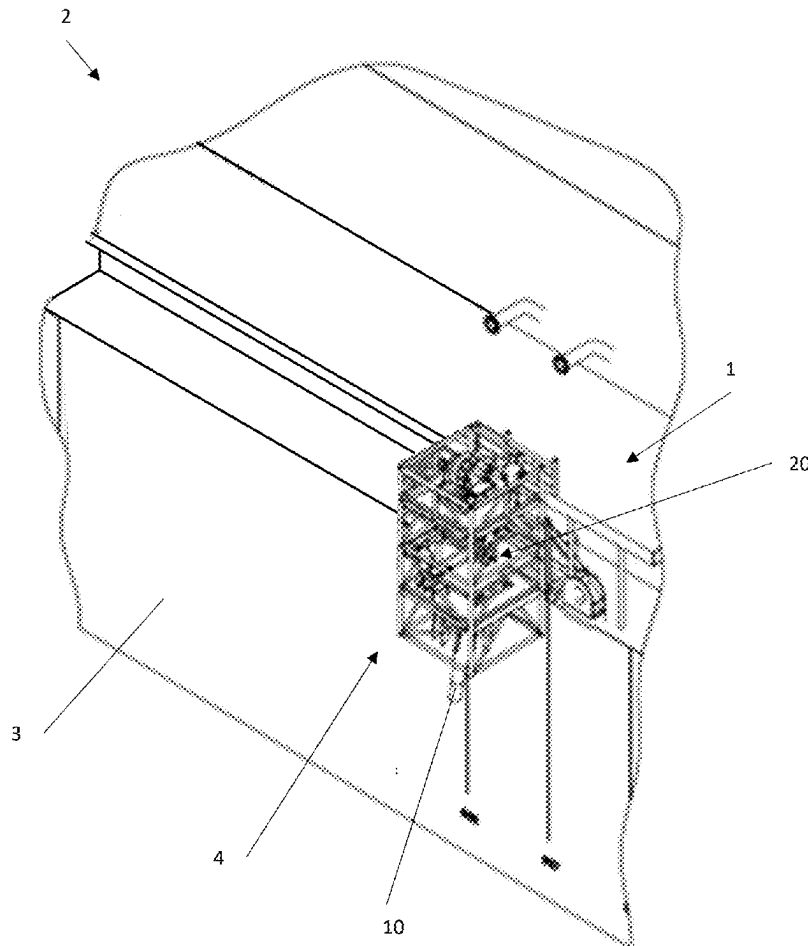
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Publication Classification(51) **Int. Cl.****B63B 27/24** (2006.01)**B63B 27/34** (2006.01)(57) **ABSTRACT**

A hang-off system is for hang-off of an end section of a first end of a transfer pipe to a floating structure or vessel. The end section of the transfer pipe is provided with a pipe hang-off flange. The hang-off system includes a gas balcony structure configured for being fastened to an outside of a shipside of the floating structure or vessel. The gas balcony structure includes a hang-off collar configured to secure the transfer pipe to the gas balcony structure. The hang-off collar includes at least two collar flanges. The at least two collar flanges define a central opening of the hang-off collar for passing through the end section of the first end of the transfer pipe. At least one of the collar flanges is movably arranged on the hang-off collar. A linear actuator moves the at least one movable collar flange between a first position and a second position. When in the first position, the central opening is larger than a transverse cross-sectional area of the pipe hang-off flange. When in the second position, the central opening is smaller than a transverse cross-sectional area of the pipe hang-off flange. A clamping mechanism is configured to clamp the pipe hang-off flange to the collar flanges.



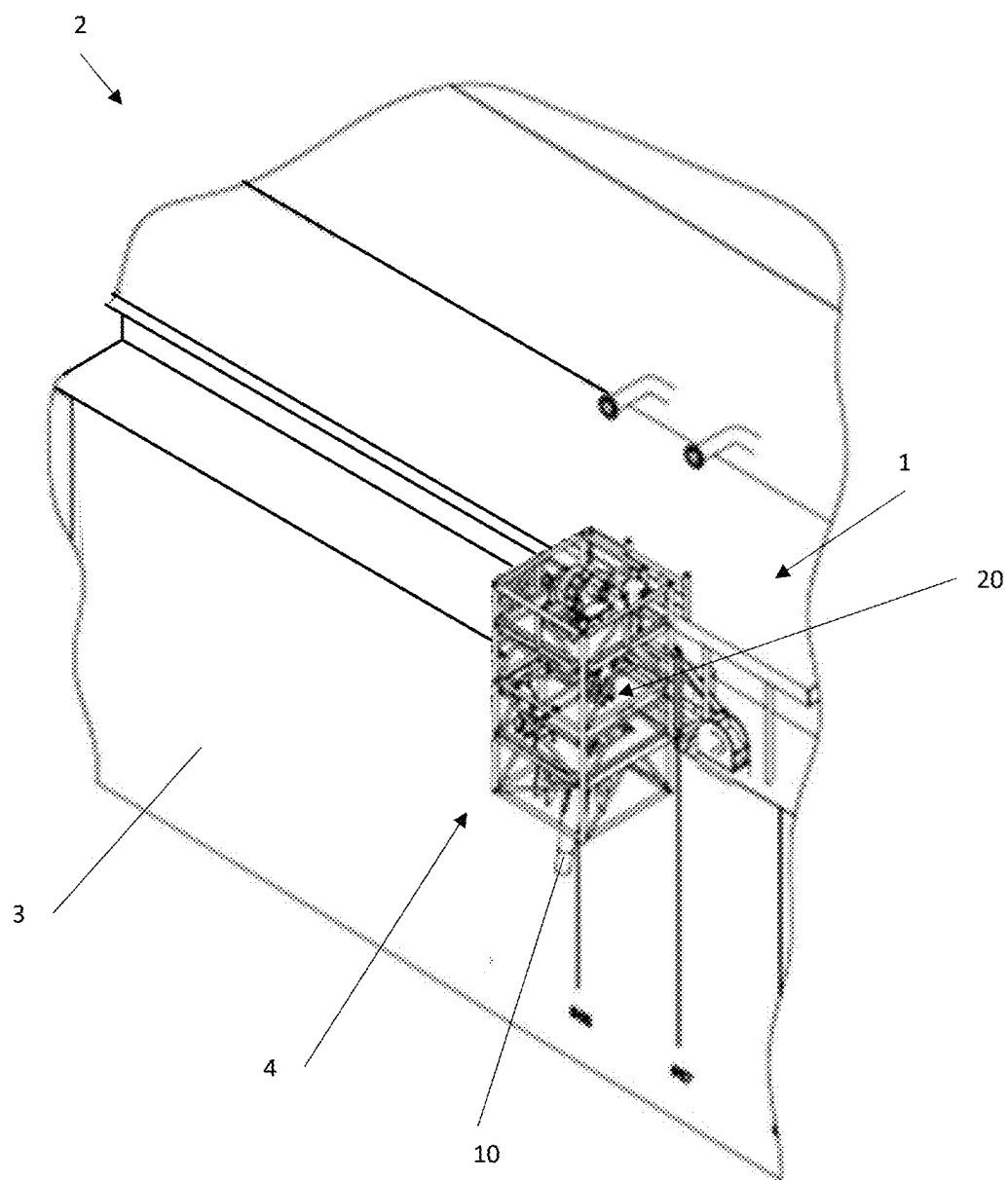


Fig. 1

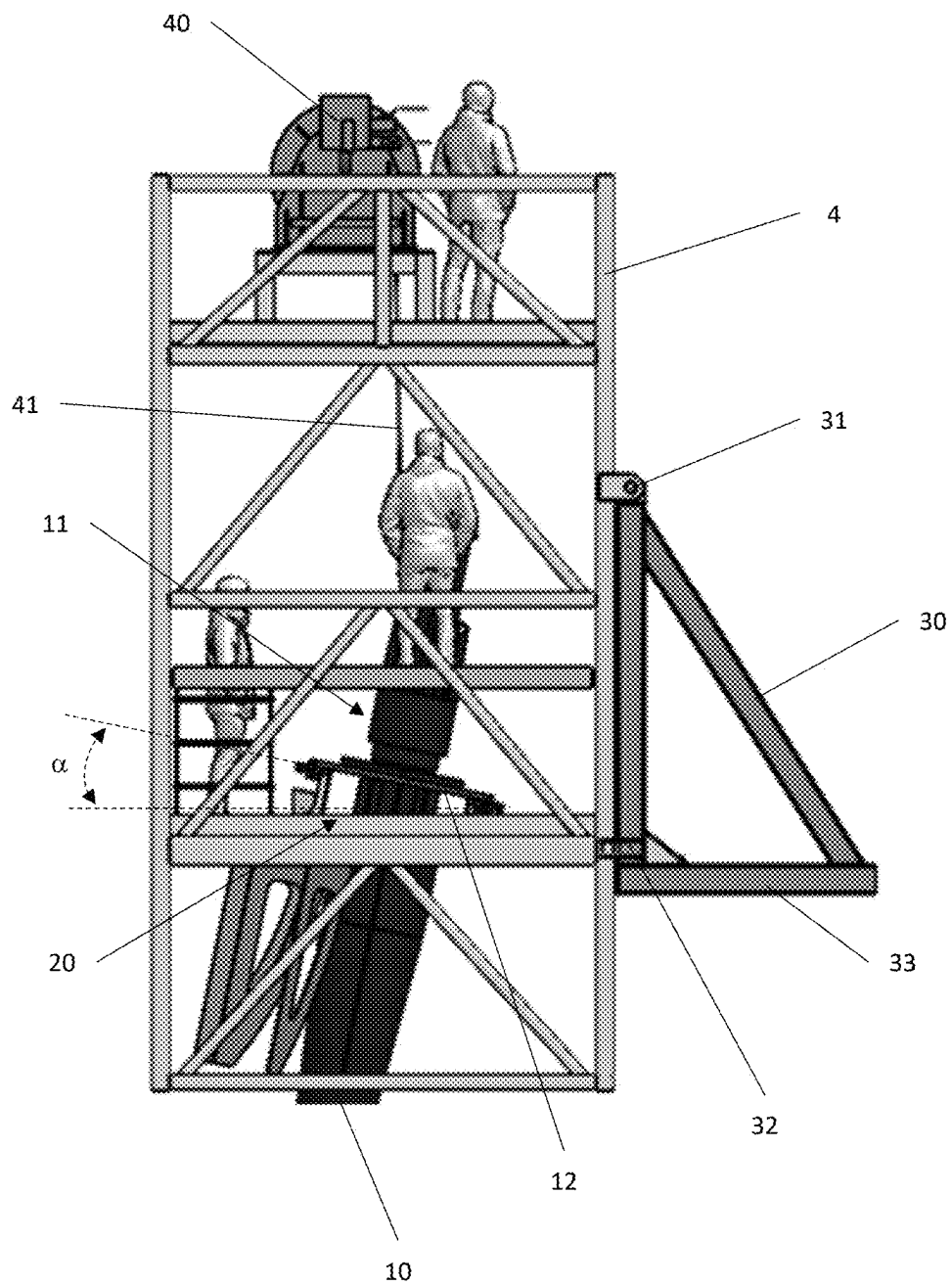


Fig. 2

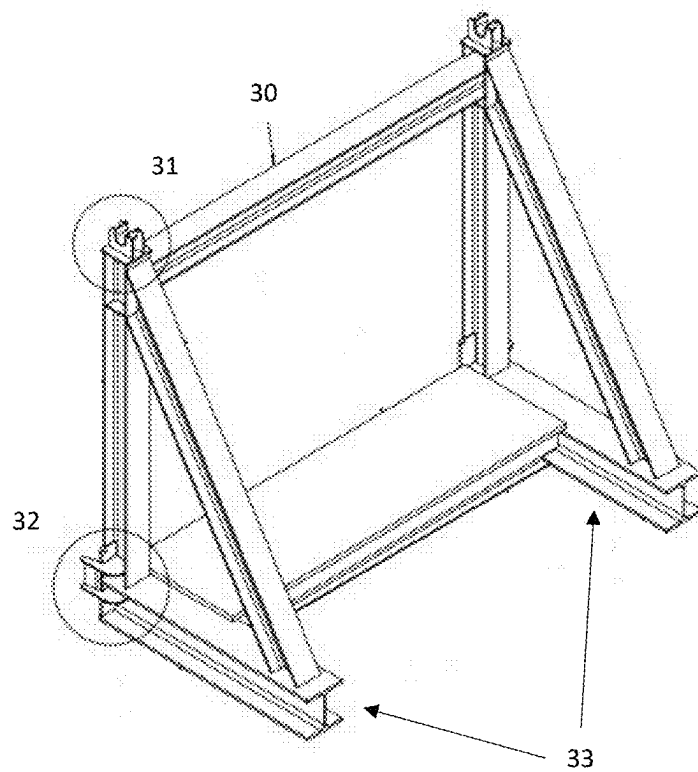


Fig. 3

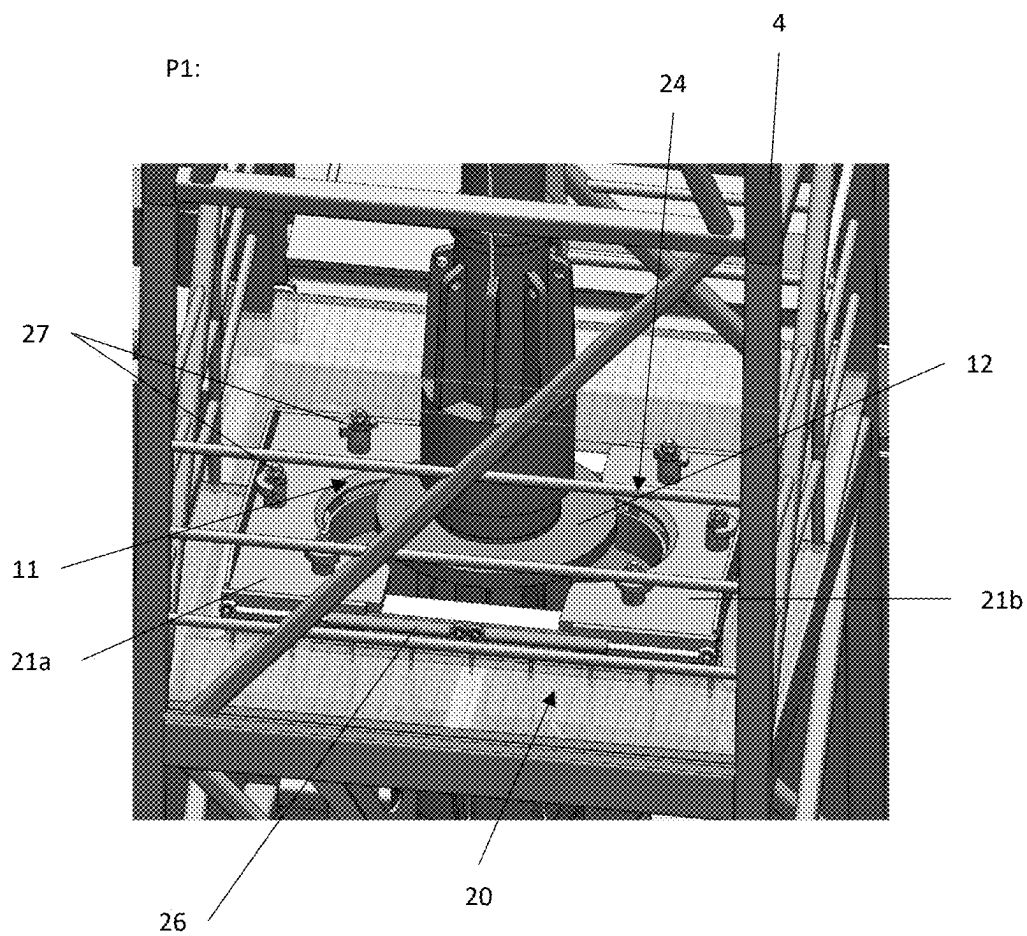
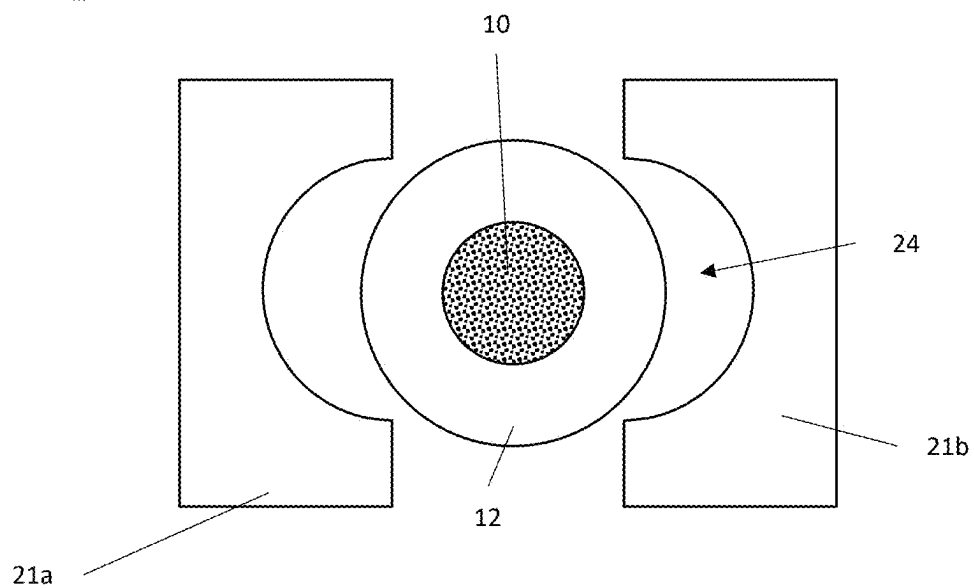


Fig. 4

P1:



P2:

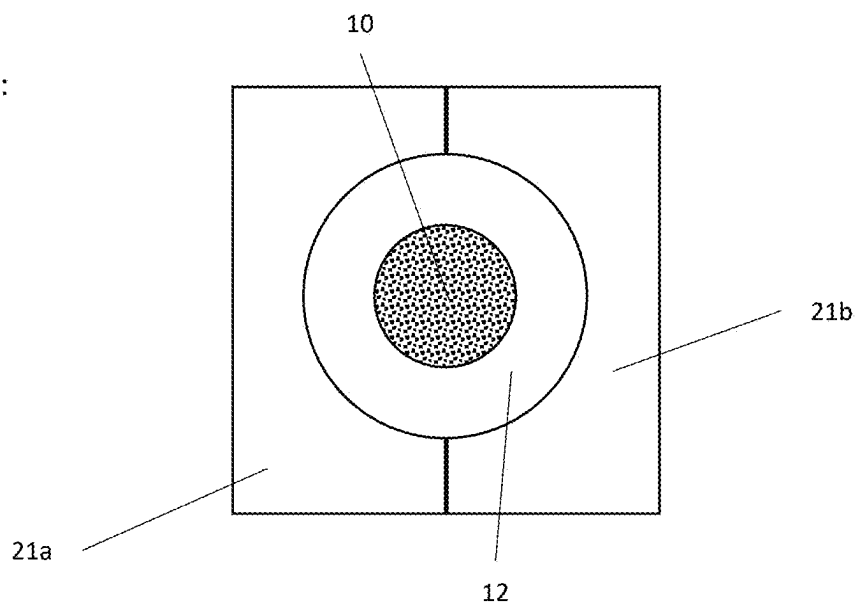


Fig. 5

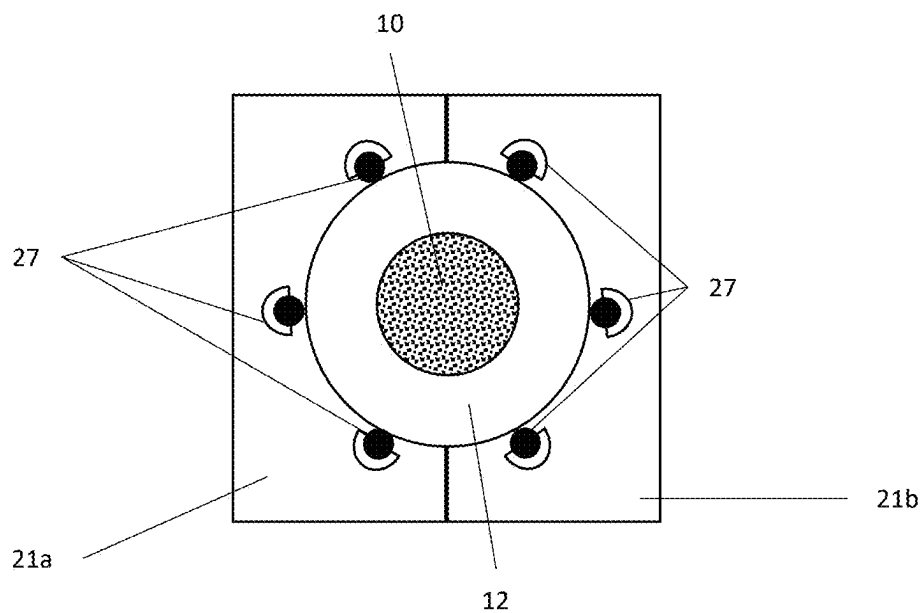


Fig. 6a

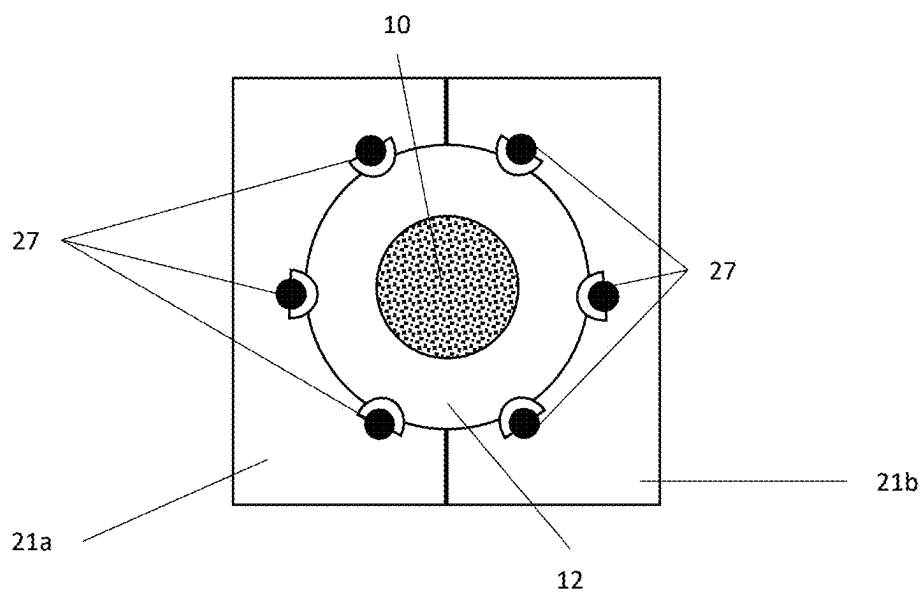


Fig. 6b

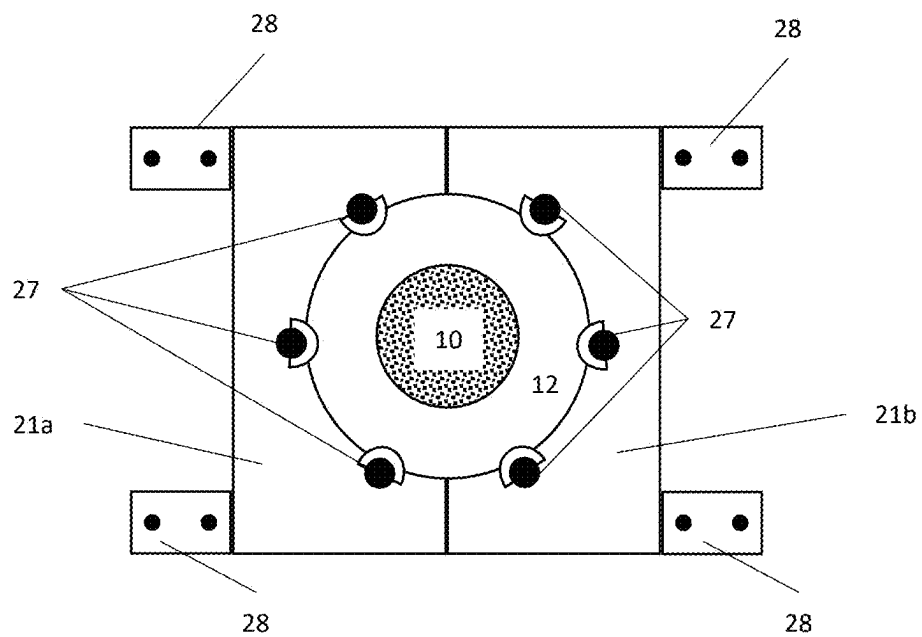


Fig. 7

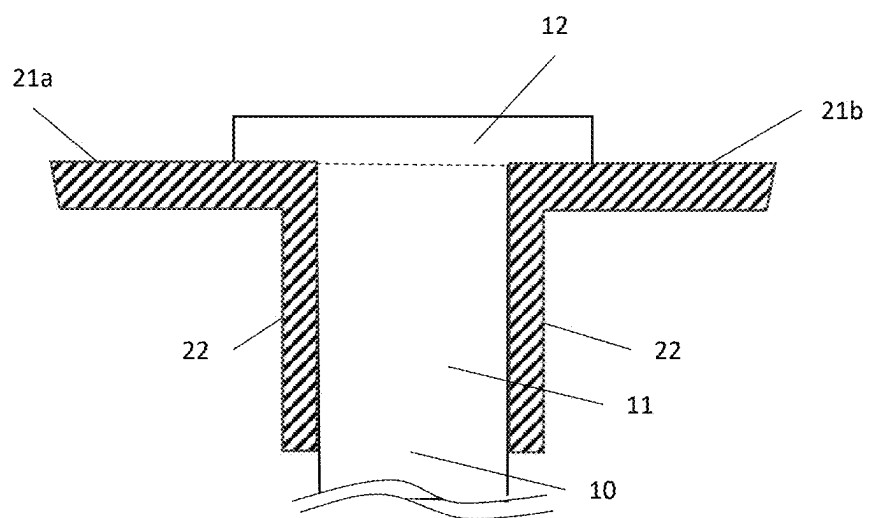


Fig. 8

HANG-OFF SYSTEM AND METHOD FOR USE OF HANG-OFF SYSTEM

TECHNICAL FIELD

[0001] The present invention relates to a hang-off system for hanging-off of a transfer pipe from a floating structure or vessel. The invention also relates to the method of pulling-in of the transfer pipe using the hang-off system.

BACKGROUND

[0002] Natural gas is used to reduce emissions locally compared to other hydrocarbon fuels. There is an increasing demand for gas worldwide. Furthermore, trade patterns are changing with new production fields emerging as well new markets access, as for example in Europe.

[0003] All these global demands require efficient transport and transfer of natural gas in the form of liquefied natural gas (LNG).

[0004] Today's loading and unloading of LNG is done via jetties, quays or other fixed infrastructure, requiring large infrastructure investments and long lead times to realize new LNG terminals.

[0005] One object of the present invention is to provide an improved hang-off system. The improved hang-off system is an enabler for floating or submerged transfer systems and thus reduced investment cost.

[0006] The present invention is suitable for LNG but can also be used for a variety of other fluids.

SUMMARY OF THE INVENTION

[0007] The present invention relates to a hang-off system for hang-off of an end section of a first end of a transfer pipe to a floating structure or vessel, the end section of the transfer pipe being provided with a pipe hang-off flange, wherein the hang-off system comprises:

[0008] a gas balcony structure configured for being fastened to an outside of a shipside of the floating structure or vessel, the gas balcony structure comprising a hang-off collar configured to secure the transfer pipe to the gas balcony structure, wherein the hang-off collar comprises:

[0009] at least two collar flanges, the at least two collar flanges defining a central opening of the hang-off collar for passing through the end section of the first end of the transfer pipe, and wherein at least one of the collar flanges is movably arranged on the hang-off collar,

[0010] a linear actuator for moving the at least one movable collar flange between a first position and a second position, wherein when in the first position, the central opening is larger than a transverse cross-sectional area of the pipe hang-off flange, and wherein when in the second position, the central opening is smaller than a transverse cross-sectional area of the pipe hang-off flange,

[0011] a clamping mechanism configured to clamp the pipe hang-off flange to the collar flanges.

[0012] In one aspect, the end section of the first end of the transfer pipe comprises a coupling for connecting to an end of a hose provided on the floating structure or vessel.

[0013] In one aspect, the hang-off collar further comprises a locking mechanism for locking the collar flanges in the second position.

[0014] In one aspect, the linear actuator is one from the group consisting of: a hydraulic, a pneumatic, and an electrical actuator.

[0015] In one aspect, the gas balcony structure comprises a winch system for pull-in and/or release of the transfer pipe through the central opening.

[0016] In one aspect, the collar flanges are arranged with an inclination towards the shipside relative to a horizontal plane of the floating structure or vessel, the inclination being adapted to a catenary profile of the transfer pipe.

[0017] In one aspect, the collar flanges are provided with pipe grips adapted to the circumference of the transfer pipe end termination for gripping around the transfer pipe when the collar flanges are in the second position and for supporting transversal or bending loads of the transfer pipe.

[0018] In one aspect, the hang-off collar comprises two or four movable collar flanges.

[0019] The present invention relates further to a method for pull-in of a first end of a transfer pipe using the hang-off system above, comprising:

[0020] having the collar flanges in the first position,

[0021] fastening a winch wire of the winch system to the end section of the transfer pipe,

[0022] hoisting the end section through the opening of the hang-off collar until the pipe hang-off flange is above the collar flanges,

[0023] when the hang-off flange is above the collar flanges, moving the collar flanges to the second position using the linear actuator and maintaining the second position, then

[0024] paying out the winch wire to lower the transfer pipe until the pipe-hang-off flange reaches the collar flanges, then

[0025] clamping the pipe hang-off flange to the collar flanges using the clamping mechanism.

[0026] In one aspect, the method further comprises locking the collar flanges in the second position using the above locking mechanism and releasing the linear actuator.

[0027] The above hang-off device enables tie-in of a flexible pipe end termination to a ship side near the midship manifold of a vessel in a safe and efficient manner. One advantage over prior art is that it can be installed on many different types of vessels with minimal impact on the hull structure. In addition, the hang-off device enables pull-in, securing, and release of a flexible pipe end termination in a safe, quick, and cost-efficient manner, with minimal impact on existing ship structure

BRIEF DESCRIPTION OF DRAWINGS

[0028] FIG. 1 illustrates schematically the layout of the hang-off system at a ship side.

[0029] FIG. 2 illustrates a side view of a gas balcony and a wire pull operation.

[0030] FIG. 3 illustrates a perspective view of a hang-off frame.

[0031] FIG. 4 illustrates a perspective view of a hang-off collar mounted within the gas balcony.

[0032] FIG. 5 shows a schematic plan view of a hang-off collar in the first and second position.

[0033] FIG. 6 shows a schematic plan view of a hang-off collar, before (a) and after (b) locking a pipe hang-off flange with a clamping mechanism.

[0034] FIG. 7 shows a schematic plan view of a hang-off collar with installed locking mechanism.

[0035] FIG. 8 shows a schematic cross section of a hang-off collar with pipe grips gripping the transfer pipe end section.

DETAILED DESCRIPTION

The Hang-Off System

[0036] A transfer system for high pressure gas applications and LNG for semi-permanent deployment, such as in floating storage and regasification unit (FSRU) or floating storage unit (FSU) applications is disclosed.

[0037] The system consists of a flexible hose reel which can reel out the hose to an FSU located less than 200 m away of the reel. The structure on the FSU side has removable gas balconies that are located on the outside of the ship side of the FSU and includes equipment for pull-in and release the flexible hose head.

[0038] The structure is divided in to three main components:

- [0039] 1) Ship foundation
- [0040] 2) Hang-off Frame
- [0041] 3) Gas Balcony Structure

[0042] The ship foundation is integrated into the ship and welded in place.

[0043] The hang-off frame is a loose structure that shall be welded to the ship foundation after final alignment of the final positions of the gas balconies in the longitudinal direction of the ship. The function of these is to give the gas balcony some good interface points that is not protruding outside of the ship side in any significant way.

[0044] Then the gas balcony structure may be lifted in place using either separate quay or supply ship crane, or by using the manifold crane to lift the gas balcony structure onto the hang-off frame. This can be done inshore or offshore in good weather conditions.

[0045] The arrangement has two platform decks, one lower and one upper with access to the winch operations. In addition, there is one foldable platform for use after the flexible pipe end termination is secured to do the tie-in operation between the spool pieces and the flexible pipe end termination.

[0046] The hang-off collar structure consists of two longitudinal flanges that are skidded with use of hydraulic cylinders. On top of these collar plates there are fixed clamping tools (for instance camlocks) to ensure that the flexible pipe end termination is fixed to the structure for bending moment.

[0047] The end termination can be pulled in by the winch through the open hang-off collar. Once the flexible pipe end termination load flange is pulled up above the hang-off collar plates, the hang-off collar is closed, to secure the end termination in position. Then it is safe for the operator to move closer and lock the camlock bolts, which will fixate the end termination to the hang-off collar. Then the pressure applied by the hydraulic cylinders can be released. When the flexible pipe end termination is safely resting on the collar plate, cam locking bolts are to be secured such that the flexible pipe end termination is resisting bending moments from the flexible pipe.

[0048] There are also 4 securing plates to secure the two collar plates in place after pull-in is complete.

[0049] The collar plates have low friction Teflon plates mounted below to reduce the friction between the flange and the collar plate during skidding operation.

[0050] This remotely operated and actuated hang-off collar arrangement allows the operator to bring the flexible pipe end termination home, lock it in place and secure it, without risk for health and safety due to moving parts.

[0051] During pull-in, the flexible pipe end termination will be guided by finger plates. These will be fixed below the collar skid plate foundation and will also have some guide up within the foundation to ensure good alignment of the flexible pipe end termination when pulled through.

A Transfer Pipe

[0052] The transfer pipe 10 may be referred to as a hose, a flexible pipe, a metallic pipe, a composite pipe or a thermoplastic composite pipe (TCP) etc. The transfer pipe itself is considered known in the art and will therefore not be described further in detail herein.

[0053] As shown in FIG. 2, the transfer pipe has an end section 11 of a first end of the 20 transfer pipe 10. The end section 11 is sometimes referred to as end termination. The end section 11 of the transfer pipe 10 is also equipped with a pipe hang-off flange 12. The purpose of this pipe hang-off flange 12 is to provide a mechanical means of hanging-off the transfer pipe 10 from a ship side 3.

[0054] Moreover, the end section 11 of the first end of the transfer pipe 10 could comprise a dry break coupling (not shown) for connecting the transfer pipe 10 to a hose provided on the floating structure or vessel 2 to provide fluid communication from the transfer pipe 10 to the floating structure or vessel 2.

A Gas Balcony Structure and a Hang-Off Frame

[0055] The hang-off system 1 comprises also a hang-off system support structure with an outside part and an inside part. The outside part of the hang-off support structure is referred here as a gas balcony structure 4 while the inside part of the hang-off system support structure is referred to as a hang-off frame 30. As shown in FIG. 2, the gas balcony 4 is typically attached to the hang-off frame 30 by an upper connector 31 and a lower connector 32. The upper and lower connectors 31, 32 35 ensure that the gas balcony structure 4 is aligned and fastened to the outside of the ship side 3. The lower member 33 of the hang-off frame 30 is integrated into the structure of the floating structure or vessel 2 and is typically fastened to a deck by welding, but can also be bolted to a foundation on the floating structure or vessel 2. The structure of the gas balcony structure 4 is a welded steel truss structure.

[0056] As shown in FIG. 2, the gas balcony structure 4 contains a hang-off collar 20 for securing the transfer pipe 10 to the gas balcony structure 4. Around the hang-off collar 20, a lower platform is provided. Moreover, the gas balcony structure 4 also contains a winch system 40 on a platform above the hang off collar 20 and the lower platform. As indicated in FIG. 2, personnel may walk on the lower platform and on the upper platform. The winch system 40, and its winch wire 41 are used to pull in or release the transfer pipe 10 relative the balcony during the operation of connecting the transfer pipe 10 to or releasing the transfer pipe 10 from the hang-off collar 20. The transfer pipe 10 enters or leaves the hang-off collar 20 within the gas balcony structure 4, through the central opening 24 of the hang-off collar 20 as shown in FIG. 4.

The Hang-Off Collar

[0057] The above hang-off collar **20** installed inside the gas balcony structure **4** as shown in FIG. **4** will now be described in detail. The hang-off collar **20** comprises two collar flanges **21a**, **21b**. The at least two collar flanges **21a**, **21b** define a central opening **24** of the hang-off collar **20** for passing through the end section **11** of the first end of the transfer pipe **10**.

[0058] The collar flanges **21a**, **21b** of the hang-off collar **20** of the hang-off system **1** are arranged with an inclination towards a shipside **3** of the floating structure or vessel **2** relative to a horizontal plane. The hang-off collar **20** is inclined by an angle α towards the shipside **3** when seen from the balcony, as illustrated in FIG. **2**. The inclination is chosen to promote a catenary configuration of the transfer pipe **10**.

[0059] The two collar flanges **21a**, **21b** shown in FIG. **4** can be moved by one or more linear actuators **26** in a plane that is perpendicular to the centerline of the transfer pipe **10**. The actuator **26** is either a hydraulic, a pneumatic, or an electrical actuator or gears or wire/winch.

[0060] The movable collar flanges **21a**, **21b** have two end positions hereby defining a first position P1 and a second position P2, as shown in FIG. **5**. The first position P1, shown in FIG. **4** and FIG. **5**, is an open position, where the central opening **24** is larger than a transverse cross-sectional area of the pipe hang-off flange **12**, hereby allowing passage of the transfer pipe **10** through the central opening **24**. The second position P2 is a locking position, where the central opening **24** is smaller than the transverse cross-sectional area of the pipe hang-off flange **12**, hereby supporting vertically the transfer pipe **10** and preventing passage of the transfer pipe **10** through the central opening **24**.

[0061] When the collar flanges **21a**, **21b** are in the second position P2, the clamping mechanism **27** shown in FIGS. **4** and **6a**, is tightened to clamp the pipe hang-off flange **12** to the surface of the collar flanges **21a**, **21b**, see also FIG. **6b**. This ensure that the transfer pipe **10** is secured against bending, due for example to the wave movement or sea currents. The clamping mechanism **27** is typically a camlock bolt. These can be bolts through aligned holes in the hang-off collar **20** and gas balcony structure **4** or plates or locking pins or locking pins in the linear actuators **26**.

[0062] Once the transfer pipe **10** is hung-off from the floating structure or vessel **2**, a locking mechanism **28**, such as the one shown in FIG. **7**, is installed on the collar flanges **21a**, **21b** to fixate the collar flanges **21a**, **21b** in the second position P2. At the same time the actuator **26** can be released so that the actuator **26** is not exerting any force onto the collar flanges **21a**, **21b**.

[0063] In another embodiment the hang-off collar **20** comprises one or more, as for example four collar flanges.

[0064] In a further embodiment, shown in FIG. **8**, the collar flanges **21a**, **21b** are provided with pipe grips **22** adapted to the circumference of the transfer pipe end termination **11** for gripping around the transfer pipe **10** when the collar flanges **21a**, **21b** are in the second position P2 and for supporting transversal or bending loads of the transfer pipe **10**.

Operating the Hang-Off System

[0065] With reference to FIGS. **2** and **4-7**, the method of pulling in a transfer pipe **10** to a ship side **3** will be described

in detail. Initially, the collar flanges **21a**, **21b** are moved to the first position P1. The winch wire **41** of the winch system **40** is lowered through the opening **24** and further down to a position below the gas balcony structure **4**, where the winch wire **41** is connected to the end section **11** of the first end of the transfer pipe **10**. The end section **11** is hoisted through the opening **24** of the hang-off collar **20** by means of the winch system **40** until the pipe hang-off flange **12** is above the collar flanges **21a**, **21b**.

[0066] Next, the collar flanges **21a**, **21b** are moved to the second position P2 using the linear actuator(s) **26**. This is followed by lowering the transfer pipe **10** by paying out the winch wire **41** until the pipe hang-off flange **12** touches the surface of the collar flanges **21a**, **21b**. The transfer pipe **10** is locked into place by means of the clamping mechanism **27** that acts on the pipe hang-off flange **12**.

[0067] Subsequently, the additional locking mechanism **28** is installed and the force in the actuator(s) **26** is released. Thus, the collar clamps **21a**, **21b** are held in place only by the locking mechanism(s) **28**.

[0068] After that, fluid communication between the transfer pipe **10** and the floating structure of vessel **2** can be established, by connecting a flexible or rigid pipe segment between the top of the transfer pipe **10** end section, and the vessel manifold, for instance by means of a combination of rigid pipe spools, a hose, and a dry break coupling.

LIST OF REFERENCE NUMBERS

[0069]	1—hang-off system
[0070]	2—floating structure or vessel
[0071]	3—shipside
[0072]	4—gas balcony structure
[0073]	10—transfer pipe
[0074]	11—end section of the first end of the transfer pipe 10
[0075]	12—pipe hang-off flange
[0076]	20—hang off collar
[0077]	21a, 21b—collar flange
[0078]	22—pipe grips of the collar flanges (21a, 21b)
[0079]	24—central opening
[0080]	26—linear actuator
[0081]	27—clamping mechanism
[0082]	28—locking mechanism
[0083]	30—hang-off frame
[0084]	31—upper connector
[0085]	32—lower connector
[0086]	33—lower member of hang-off frame
[0087]	40—winch system
[0088]	41—winch wire
[0089]	P1—first position of the movable collar flange
[0090]	P2—second position of the movable collar flange

1. A hang-off system for hang-off of an end section of a first end of a transfer pipe to a floating structure or vessel, the end section of the transfer pipe being provided with a pipe hang-off flange,

wherein the hang-off system comprises:

a gas balcony structure configured for being fastened to an outside of a shipside of the floating structure or vessel, the gas balcony structure comprising a hang-off collar configured to secure the transfer pipe to the gas balcony structure,

wherein the hang-off collar comprises:

at least two collar flanges, the at least two collar flanges defining a central opening of the hang-off collar for passing through the end section of the first end of the transfer pipe, and wherein at least one of the collar flanges is movably arranged on the hang-off collar,

a linear actuator for moving the at least one movable collar flange between a first position and a second position, wherein when in the first position, the central opening is larger than a transverse cross-sectional area of the pipe hang-off flange, and wherein when in the second position, the central opening is smaller than a transverse cross-sectional area of the pipe hang-off flange,

a clamping mechanism configured to clamp the pipe hang-off flange to the collar flanges.

2. The hang-off system according to claim 1, wherein the end section of the first end of the transfer pipe comprises a coupling for connecting to an end of a hose provided on the floating structure or vessel.

3. The hang-off system according to claim 1, wherein the hang-off collar further comprises a locking mechanism for locking the collar flanges in the second position.

4. The hang-off system according to claim 1, wherein the linear actuator is one from the group consisting of: a hydraulic, a pneumatic, and an electrical actuator.

5. The hang-off system according to claim 1, wherein the gas balcony structure comprises a winch system for pull-in and/or release of the transfer pipe through the central opening.

6. The hang-off system according to claim 1, wherein the collar flanges are arranged with an inclination towards the shipside relative to a horizontal plane of the floating structure or vessel, the inclination being adapted to a catenary profile of the transfer pipe.

7. The hang-off system according to claim 1, wherein the collar flanges are provided with pipe grips adapted to the circumference of the transfer pipe end termination for gripping around the transfer pipe when the collar flanges are in the second position and for supporting transversal or bending loads of the transfer pipe.

8. The hang-off system according to claim 1, wherein the hang-off collar comprises two or four movable collar flanges.

9. A method for pull-in of a first end of a transfer pipe using a hang-off system for hang-off of an end section of a first end of a transfer pipe to a floating structure or vessel, the end section of the transfer pipe being provided with a pipe hang-off flange,

wherein the hang-off system comprises:

a gas balcony structure configured for being fastened to an outside of a shipside of the floating structure or vessel, the gas balcony structure comprising a hang-off collar configured to secure the transfer pipe to the gas balcony structure,

wherein the hang-off collar comprises:

at least two collar flanges, the at least two collar flanges defining a central opening of the hang-off collar for passing through the end section of the first end of the transfer pipe, and wherein at least one of the collar flanges is movably arranged on the hang-off collar,

a linear actuator for moving the at least one movable collar flange between a first position and a second position, wherein when in the first position, the central opening is larger than a transverse cross-sectional area of the pipe hang-off flange, and wherein when in the second position, the central opening is smaller than a transverse cross-sectional area of the pipe hang-off flange,

a clamping mechanism configured to clamp the pipe hang-off flange to the collar flanges,

wherein the gas balcony structure comprises a winch system for pull-in and/or release of the transfer pipe through the central opening, the method comprising:

having the collar flanges in the first position,

fastening a winch wire of the winch system to the end section of the transfer pipe,

hoisting the end section through the opening of the hang-off collar until the pipe hang-off flange is above the collar flanges,

when the hang-off flange is above the collar flanges, moving the collar flanges to the second position using the linear actuator and maintaining the second position, then

paying out the winch wire to lower the transfer pipe until the pipe-hang-off flange reaches the collar flanges, then clamping the pipe hang-off flange to the collar flanges using the clamping mechanism.

10. The method according to claim 9, wherein the hang-off collar further comprises a locking mechanism for locking the collar flanges in the second position, the method further comprising locking the collar flanges in the second position using the locking mechanism, and releasing the linear actuator.

* * * * *